CHAPTER 6

Divisibility and congruences

Divisibility in $\mathbb{Z}$, Euclidean division, congruences

Objectives. Use divisibility in $\mathbb{Z}$ and its properties; carry out a Euclidean division; compute with congruences; establish a divisibility test; study remainders of powers.

Prerequisites. Multiples, divisors, prime numbers (Seconde); proof by induction; integer part of a real number.

Fluency

Fluency · No calculator, five minutes.

1. List the positive divisors of 36.
2. Is the number $n(n+1)$ always even? Justify.
3. What is the units digit of 7^4?
4. Simplify the fraction $\frac{84}{126}$.
5. Is the number 2,026 divisible by 3? by 2?
6. Compute 17×23.

Solution · Fluency.

1. 1, 2, 3, 4, 6, 9, 12, 18, 36.
2. Yes: one of two consecutive integers is even.
3. $7^2 = 49$, $7^4 = 2{,}401$: the units digit is 1.
4. $\frac{84}{126} = \frac{2\times 42}{3\times 42} = \frac{2}{3}$.
5. The sum of the digits is 10, not a multiple of 3: no. It is even: yes.
6. 391.

Introductory activity

Activity 1 · What day will it be? 35 min . 17 October 2026 is a Saturday. The days of the week are numbered from 0 (Saturday) to 6 (Friday): Sunday has number 1, Monday number 2, and so on.

1. What day of the week will it be 7 days later? 14 days? 700 days?
2. What day will it be 100 days later? Explain how to get the answer without counting the days one by one.
3. Same question for 1,000 days.
4. The year 2027 is not a leap year. What day of the week will 17 October 2027 be?
5. 1 January 2027 comes 76 days after 17 October 2026. What day will it be?
6. Two durations a and b, in days, lead to the same day of the week. What can be said about $a - b$? State a general condition.

Solution · Activity.

1. Saturday in all three cases, because 7, 14 and 700 are multiples of 7.
2. $100 = 7 \times 14 + 2$: we move forward 2 days, it is a Monday. Only the remainder of the division by 7 matters.
3. $1{,}000 = 7 \times 142 + 6$: Friday.
4. $365 = 7 \times 52 + 1$: Sunday.
5. $76 = 7 \times 10 + 6$: Friday.
6. a and b have the same remainder in the division by 7, that is $a - b$ is a multiple of 7. We shall say that a and b are congruent modulo 7.

Teacher note · Running the activity. Have the answer to question 4 checked with a calendar or a spreadsheet, pointing out that checking an example does not replace the reasoning. Extension: explain why 17 October 2030 is a Thursday (1,461 days, including one 29 February, and $1{,}461 = 7 \times 208 + 5$).

Course notes

Divisibility in $\mathbb{Z}$

Definition · Divisor, multiple. Let a and b be two integers. We say that a *divides* b, written $a \mid b$, if there is an integer k such that $b = ka$. We also say that a is a *divisor* of b, or that b is a *multiple* of a.

Property · Properties of divisibility. Let a, b, c be integers.

1. If $a \mid b$ and $b \mid c$, then $a \mid c$.
2. If $a \mid b$ and $a \mid c$, then $a \mid bu + cv$ for all integers u and v.
3. If $a \mid b$ and $b \neq 0$, then $|a| \leq |b|$.

Proof. 1. $b = ka$ and $c = k'b$ give $c = (k'k)a$. 2. $b = ka$ and $c = k'a$ give $bu + cv = (ku + k'v)a$. 3. $b = ka$ with $k \neq 0$, so $|k| \geq 1$ and $|b| = |k|\,|a| \geq |a|$. □

Property · Euclidean division. Let $a \in \mathbb{Z}$ and $b \in \mathbb{N}$, $b \geq 1$. There is a unique pair of integers (q, r) such that $a = bq + r$ and $0 \leq r < b$. We call q the *quotient* and r the *remainder*.

Proof. *Existence.* Let $q = \left\lfloor \frac{a}{b} \right\rfloor$, the largest integer less than or equal to $\frac{a}{b}$. Then $q \leq \frac{a}{b} < q + 1$, so $bq \leq a < bq + b$, and $r = a - bq$ satisfies $0 \leq r < b$.

Uniqueness. If $a = bq + r = bq' + r'$ with $0 \leq r, r' < b$, then $b(q - q') = r' - r$ and $-b < r' - r < b$. The only multiple of b strictly between $-b$ and b is 0, so $r = r'$ and then $q = q'$. □

Watch out. The remainder is always nonnegative: $-47 = 5 \times (-10) + 3$, so the division of -47 by 5 has quotient -10 and remainder 3, not -9 and -2.

Congruences

Definition · Congruence modulo n. Let $n \geq 2$ be an integer. Two integers a and b are *congruent modulo* n, written $a \equiv b \pmod{n}$, if $n \mid a - b$.

Property · Congruence and remainder. $a \equiv b \pmod{n}$ if and only if a and b have the same remainder in the Euclidean division by n. In particular, every integer is congruent modulo n to its remainder, which belongs to $\{0, \ldots, n-1\}$.

Proof. Write $a = nq + r$ and $b = nq' + r'$ with $0 \leq r, r' < n$. Then $a - b = n(q - q') + (r - r')$ and $-n < r - r' < n$. So $n \mid a - b$ if and only if $n \mid r - r'$, if and only if $r = r'$. □

Property · Compatibility with the operations. If $a \equiv b \pmod{n}$ and $c \equiv d \pmod{n}$, then

$$a + c \equiv b + d, \qquad a - c \equiv b - d, \qquad ac \equiv bd, \qquad a^k \equiv b^k \pmod{n}$$

for every natural number k. The relation is also transitive: if $a \equiv b$ and $b \equiv c \pmod{n}$, then $a \equiv c \pmod{n}$.

Proof. Sum: $(a + c) - (b + d) = (a - b) + (c - d)$ is a sum of two multiples of n. Product: $ac - bd = a(c - d) + d(a - b)$, a multiple of n. Power: induction on k from the product. Transitivity: $a - c = (a - b) + (b - c)$. □

Watch out. A congruence is not simplified like an equality: $2 \times 3 \equiv 2 \times 1 \pmod 4$, but $3 \not\equiv 1 \pmod 4$. Division is possible only in specific cases (chapter 7).

Method · Remainder of a large power. Look for a power congruent to 1 (or to -1), then use the Euclidean division of the exponent. Example: $2^3 = 8 \equiv 1 \pmod 7$ and $2{,}026 = 3 \times 675 + 1$, so $2^{2026} = \left(2^3\right)^{675} \times 2 \equiv 1^{675} \times 2 = 2 \pmod 7$.

Method · Table of congruences. To study an expression modulo n, draw up a table giving its value modulo n for each possible remainder of x. Example: modulo 4, $x \equiv 0, 1, 2, 3$ gives $x^2 \equiv 0, 1, 0, 1$. A square is therefore congruent to 0 or 1 modulo 4.

Method · Divisibility test for 9. If $N = \sum_{k=0}^{p} a_k 10^k$ is written with the digits $a_0, \ldots, a_p$, since $10 \equiv 1 \pmod 9$, we have $10^k \equiv 1$ and $N \equiv a_0 + a_1 + \cdots + a_p \pmod 9$. An integer is divisible by 9 if and only if the sum of its digits is.

Exercises

Exercise 6.1 · Euclidean divisions.

1. Carry out the Euclidean division of 2,026 by 17.
2. Carry out the Euclidean division of -47 by 5, then of -35 by 4.
3. Determine the natural numbers n whose quotient in the division by 6 equals the remainder.

Solution.

1. $2{,}026 = 17 \times 119 + 3$: quotient 119, remainder 3.
2. $-47 = 5 \times (-10) + 3$ and $-35 = 4 \times (-9) + 1$.
3. $n = 6q + q = 7q$ with $0 \leq q < 6$: $n \in \{0, 7, 14, 21, 28, 35\}$.

Exercise 6.2 · Remainders of powers.

1. Determine the remainder of 3^{100} in the division by 5.
2. Determine the remainder of $5^{2026} + 2^{2026}$ in the division by 3.

Solution.

1. $3^4 = 81 \equiv 1 \pmod 5$, so $3^{100} = \left(3^4\right)^{25} \equiv 1 \pmod 5$: the remainder is 1.
2. $5 \equiv 2 \equiv -1 \pmod 3$, so $5^{2026} + 2^{2026} \equiv 2 \times (-1)^{2026} = 2 \pmod 3$: the remainder is 2.

Exercise 6.3 · Sums of two squares.

1. Prove that for every integer x, $x^2 \equiv 0$ or $x^2 \equiv 1 \pmod 4$.
2. Deduce that an integer congruent to 3 modulo 4 is not the sum of two squares of integers.
3. Can the integer 2,027 be written $x^2 + y^2$ with x, y integers?

Solution.

1. Table: $x \equiv 0, 1, 2, 3 \pmod 4$ gives $x^2 \equiv 0, 1, 4, 9$, that is $0, 1, 0, 1 \pmod 4$.
2. $x^2 + y^2$ is congruent to $0 + 0$, $0 + 1$ or $1 + 1$, hence to 0, 1 or 2 modulo 4, never to 3.
3. $2{,}027 = 4 \times 506 + 3$: no.

Exercise 6.4 · Divisibility test for 11.

1. Prove that $10 \equiv -1 \pmod{11}$, then that $10^k \equiv (-1)^k \pmod{11}$.
2. Deduce that an integer is divisible by 11 if and only if the alternating sum of its digits, starting from the units, $a_0 - a_1 + a_2 - \ldots$, is.
3. Is the integer 918,082 divisible by 11?

Solution.

1. $10-(-1)=11$. Compatibility with powers gives $10^k \equiv (-1)^k$.
2. $N=\sum a_k 10^k \equiv \sum a_k(-1)^k \pmod{11}$; N and the alternating sum have the same remainder.
3. $2-8+0-8+1-9=-22$, a multiple of 11: yes ($918{,}082 = 11 \times 83{,}462$).

Exercise 6.5 · Divisibility for every integer. Prove that for every natural number n:

1. 4^n-1 is divisible by 3;
2. n^3-n is divisible by 6.

Solution.

1. $4 \equiv 1 \pmod 3$, so $4^n \equiv 1$ and $3 \mid 4^n-1$.
2. Table modulo 6: for $n \equiv 0,1,2,3,4,5$, $n^3-n \equiv 0,0,6,24,60,120$, all multiples of 6. So $n^3-n \equiv 0 \pmod 6$ for every n. Remark: $n^3-n=(n-1)n(n+1)$ is a product of three consecutive integers; chapter 7 will show why being divisible by 2 and by 3 is enough.

Exercise 6.6 · Congruence equations.

1. Solve $3x \equiv 5 \pmod 7$. First look for an integer u such that $3u \equiv 1 \pmod 7$.
2. Solve $x^2 \equiv 2 \pmod 7$ with a table.

Solution.

1. $3 \times 5 = 15 \equiv 1 \pmod 7$. Multiplying by 5: $3x \equiv 5$ implies $x \equiv 25 \equiv 4 \pmod 7$; conversely $3 \times 4 = 12 \equiv 5$. The solutions are the integers $x = 4+7k$, $k \in \mathbb{Z}$.
2. For $x \equiv 0,1,\ldots,6 \pmod 7$: $x^2 \equiv 0,1,4,2,2,4,1$. The solutions are the integers congruent to 3 or 4 modulo 7.

Exercise 6.7 · The last two digits. Determine the last two digits of the decimal expansion of 7^{2026}.

Solution. We work modulo 100. $7^2 = 49$ and $7^4 = 2{,}401 \equiv 1 \pmod{100}$. Since $2{,}026 = 4 \times 506 + 2$, $7^{2026} \equiv 7^2 = 49 \pmod{100}$: the expansion ends in 49.

Problem

Problem · The check digit of an ISBN-13 code. An ISBN-13 code is a sequence of thirteen digits $a_1a_2\dots a_{13}$. Let

$$S = a_1 + 3a_2 + a_3 + 3a_4 + \cdots + 3a_{12} + a_{13},$$

with the weights 1 and 3 alternating. The code is valid if and only if $S \equiv 0 \pmod{10}$; the last digit a_{13}, called the check digit, is chosen to make it so.

1. Consider the fictitious number 978-2-0601-2345-a_{13}. Compute the check digit a_{13}.
2. Assume that a single digit a is replaced by a digit $a' \neq a$.
   (a) Prove that S changes by $w(a' - a)$, with $w \in \{1, 3\}$.
   (b) Check that $7 \times 3 \equiv 1 \pmod{10}$ and deduce that $3(a' - a) \equiv 0 \pmod{10}$ implies $a' - a \equiv 0 \pmod{10}$.
   (c) Conclude: an error in a single digit is always detected.
3. Assume that two distinct adjacent digits a and b are swapped.
   (a) Prove that S changes by $2(b - a)$ or by $2(a - b)$.
   (b) Determine under which condition on a and b the error is not detected. Give an example.

Solution · Problem.

1. The first twelve digits $9, 7, 8, 2, 0, 6, 0, 1, 2, 3, 4, 5$ give $9 + 21 + 8 + 6 + 0 + 18 + 0 + 3 + 2 + 9 + 4 + 15 = 95$. We need $95 + a_{13} \equiv 0 \pmod{10}$ with $0 \leq a_{13} \leq 9$: $a_{13} = 5$.
2. (a) The term wa becomes wa', the other terms do not change.
   (b) $21 \equiv 1 \pmod{10}$. If $3(a' - a) \equiv 0$, multiplying by 7: $a' - a \equiv 21(a' - a) \equiv 0 \pmod{10}$. The case $w = 1$ is immediate.
   (c) $a' - a$ is a multiple of 10 between -9 and 9, hence zero: a contradiction with $a' \neq a$. The new sum is no longer a multiple of 10, so the error is detected.
3. (a) The two digits have weights 1 and 3. The contribution $a + 3b$ becomes $b + 3a$ (or the reverse), a change of $2(a - b)$ or of $2(b - a)$.
   (b) The error is not detected if and only if $10 \mid 2(a - b)$, that is $5 \mid a - b$: $|a - b| = 5$. Example: swapping the adjacent digits 1 and 6 does not change S modulo 10.

Teacher note · Status of the problem. Check codes (ISBN, bank account keys, barcodes) are a "possible problem" of the syllabus. Question 2b uses an inverse modulo 10: the notion is formalised in chapter 7. The number in question 1 is given as an example, for the computation.